

Solving rational inequalities algebraically

Basic skills

Q1. $f(x) = \frac{2x+5}{6-x}$

a) $N(x) = 2x + 5$, $D(x) = 6 - x$

b) $N(x) = 0$ at $x = -\frac{5}{2} = -2.5$

$D(x) = 0$ at $x = 6$

c)

+ / -	$x < -2.5$	$-2.5 < x < 6$	$x > 6$
$2x + 5$	-	+	+
$x - 6$	-	-	+
$f(x)$	$\frac{-}{-} = +$	$\frac{+}{-} = -$	$\frac{+}{+} = +$

So $f(x) < 0$ for $-2.5 < x < 6$

$f(x) > 0$ for $x > 6$ or $x < -2.5$

Q2. critical pts:

$$6x - 36 = 0 \quad x = 6$$

$$2x - 4 = 0 \quad x = 2$$

+/-	$x < 2$	$2 < x < 6$	$x > 6$
$6x - 36$	-	-	+
$2x - 4$	-	+	+
$f(x)$	$\frac{-}{-} = +$	$\frac{-}{+} = -$	$\frac{+}{+} = +$

so $f(x) > 0$ for $x < 2$ or $x > 6$

$f(x) < 0$ for $2 < x < 6$

Competency skills

Q3. a) critical points

$$x^2 - 9 = 0 \quad x = \pm 3$$

$$10 - 2x = 0 \quad x = 5$$

so ranges $x < -3$, $-3 < x < 3$, $3 < x < 5$, $x > 5$

+/-	$x < -3$	$-3 < x < 3$	$3 < x < 5$	$x > 5$
$x^2 - 9$	+	-	+	+
$10 - 2x$	+	+	+	-
$f(x)$	$\frac{+}{+} = +$	$\frac{-}{+} = -$	$\frac{+}{+} = +$	$\frac{+}{-} = -$

Some people use

$(x-3), (x+3)$ as separate factors
for $N(x)$

b) so $f(x) \geq 0$ at $x = \pm 3$ by
 $N(x) = 0$

or

$$x < -3 \text{ or } 3 < x < 5$$

$$\text{so } x \leq -3 \text{ or } 3 \leq x < 5$$

c) $f(x) < 0$ for $-3 < x < 3$ or $x > 5$

Q4. a) $\frac{x+3}{(x-2)(5-x)} \leq 0$ ← could use table alternatively case by case

case by case method:

$x(x-2)$ split into ① $x-2 > 0$ and ② $x-2 < 0$

① $\frac{x+3}{5-x} \leq 0$ $x(5-x)$ split into

② $5-x > 0$

③ $5-x < 0$

1a

$x+3 \leq 0 \Rightarrow x \leq -3$

and $x > 2$ and $x < 5$

X

no sol.

sign change as $5-x < 0$

1b

$x+3 \geq 0 \Rightarrow x \geq -3$

and $x > 2$, $x > 5$

so

yes for $x > 5$

← combine inequalities

$$\textcircled{2} \quad \frac{x+3}{5-x} \geq 0 \quad \times (5-x)$$

$$\textcircled{a} \quad 5-x > 0$$

$$\textcircled{b} \quad 5-x < 0$$

$$\textcircled{2a} \quad x+3 \geq 0 \Rightarrow x \geq -3$$

$$\text{and } x < 2, \quad x < 5$$

✓ combine

$$\text{so for } -3 \leq x < 2$$

$$\textcircled{2b} \quad x+3 \leq 0 \Rightarrow x \leq -3$$

$$\text{and } x < 2, \quad x > 5$$

no sol^x

so answer is for $x > 5$ or $-3 \leq x < 2$

$$\text{b) } \frac{5}{3(x-2)} \leq \frac{8}{3(x-5)} \Rightarrow \frac{5}{3(x-2)} - \frac{8}{3(x-5)} \leq 0$$

$$\frac{5(x-5) - 8(x-2)}{3(x-2)(x-5)} \leq 0$$

$$\frac{-3x - 9}{3(x-2)(x-5)} \leq 0 \Rightarrow \frac{x+3}{(x-2)(5-x)} \leq 0$$

equiv fractions $x^{-\frac{1}{3}}$

So same answer as a)

$$x > 5 \text{ or } -3 \leq x < 2$$

Q5. $x+5 \geq \frac{1}{x}$

$$x+5 - \frac{1}{x} \geq 0$$

$$\frac{x^2 + 5x - 1}{x} \geq 0$$

table or

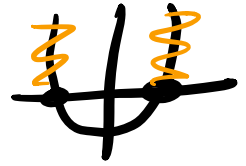
could not have x^2 to do two cases

so consider two cases $x \geq 0$
 $x < 0$

• If $x > 0$ $x^2 + 5x - 1 \geq 0$

roots of $x^2 + 5x - 1$ are

$$x = \frac{-5 \pm \sqrt{29}}{2}$$



so $\left(x \geq \frac{-5 + \sqrt{29}}{2} \text{ or } x \leq \frac{-5 - \sqrt{29}}{2} \right)$
and
 $x > 0$

so $x \geq \frac{\sqrt{29} - 5}{2}$

• If $x < 0$ $x^2 + 5x - 1 \leq 0$

so $\left(\frac{-5 - \sqrt{29}}{2} \leq x \leq \frac{\sqrt{29} - 5}{2} \right)$
and
 $x < 0$

so $-\frac{5 + \sqrt{29}}{2} \leq x < 0$

So full answer is

$$\frac{-5 - \sqrt{29}}{2} \leq x < 0 \text{ or } x \geq \frac{\sqrt{29} - 5}{2}$$

Q 6. $\frac{x+1}{x+2} > 2x+1$

$$\frac{x+1}{x+2} - 2x+1 > 0$$

$$\frac{x+1 - (2x+1)(x+2)}{x+2} > 0$$

$$\frac{-2x^2 - 4x - 1}{x+2} > 0$$

$x-1$ ✓

$$\frac{2x^2 + 4x + 1}{x+2} < 0$$

✓ $x-1$

so sign
change

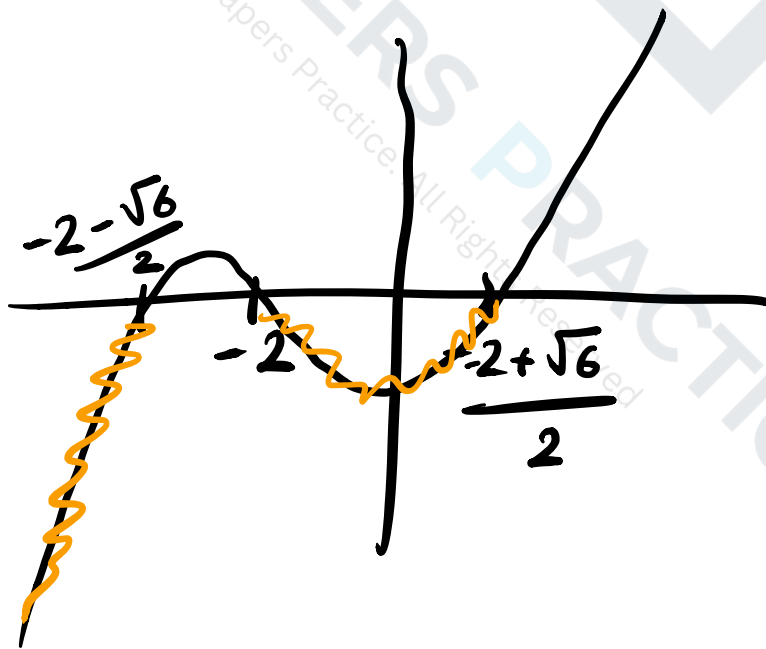
now $x(x+2)^2$ (or table or cases)

$$(2x^2 + 4x + 1)(x+2) < 0$$

for roots

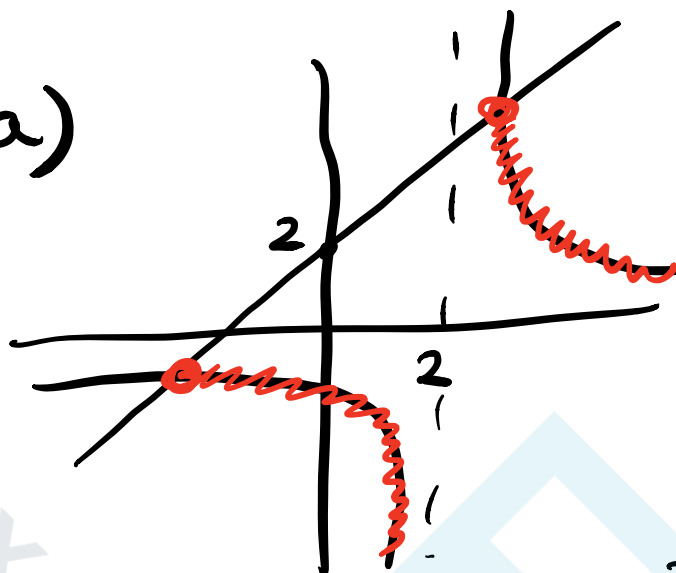
$$x = \frac{-4 \pm \sqrt{24}}{2 \times 2} = \frac{-2 \pm \sqrt{6}}{2}$$

$$\text{or } x = -2$$



$$\text{so } x < -\frac{2 - \sqrt{6}}{2} \text{ or } -2 < x < \frac{-2 + \sqrt{6}}{2}$$

Q7. a)



$$\frac{1}{x-2} = 5x+2$$

$$1 = (5x+2)(x-2)$$

$$5x^2 - 8x - 5 = 0$$

$$x = \frac{8 \pm \sqrt{64+100}}{2 \times 5}$$

$$x = \frac{8 \pm 2\sqrt{41}}{10} = \frac{4 \pm 2\sqrt{41}}{5}$$

b) on graph see where reciprocal is lower

$$\text{so } x > \frac{4+2\sqrt{41}}{5}$$

or

$$\frac{4-2\sqrt{41}}{5} < x < 2$$

Q8. $\frac{x^2 - 2x + 1}{x^2 - 6x + 8} \geq 0$

Factorise $\frac{(x-1)^2}{(x-4)(x-2)} \geq 0$

$(x-4)(x-2)$
 $\uparrow$
 $x = 2, 4$

$N(x) \neq 0$ for any x

So don't need its for table

Table

$+/-$	$x < 2$	$2 < x < 4$	$x > 4$
$N(x)$	+	+	+
$x-4$	-	-	+
$x-2$	-	+	+
$f(x)$	$\frac{+}{-x-} = \frac{+}{+} = +$	$\frac{+}{-x+} = \frac{+}{-} = -$	$\frac{+}{++} = \frac{+}{+} = +$

So $x < 2$ or $x > 4$

but $N(x) = 0$ at $x = 1$

So $x < 2$ or $x > 4$ or $x = 1$

Q9. Because multiplying by variables need careful consideration of there positivity/negativity

Q10. $\frac{5x+1}{2x} < \frac{8}{x+1}$

so $\frac{5x+1}{2x} - \frac{8}{x+1} < 0$

$$\frac{(5x+1)(x+1) - 8 \times 2x}{2x(x+1)} < 0$$

$$\frac{5x^2 + 6x + 5 - 16x}{2x(x+1)} < 0$$

$$\frac{5x^2 - 10x + 5}{2x(x+1)} < 0$$

$$\frac{x^2 - 2x + 1}{2x(x+1)} < 0$$

$$(x-1)^2 \geq 0 \text{ for any } x$$

$$\frac{(x-1)^2}{2x(x+1)} < 0$$

critical at $x=0, -1$

Table

+/-	$x < -1$	$-1 < x < 0$	$x > 0$
$(x-1)^2$	+	+	+
x	-	-	+
$x+1$	-	+	+
$f(x)$	$\frac{+}{-x-} = \frac{+}{+} = +$	$\frac{+}{-x+} = \frac{+}{-} = -$	$\frac{+}{+x+} = \frac{+}{+} = +$

so $-1 < x < 0$

Q11. As denominator is ≥ 0 (in fact > 5)
so can multiply without problem

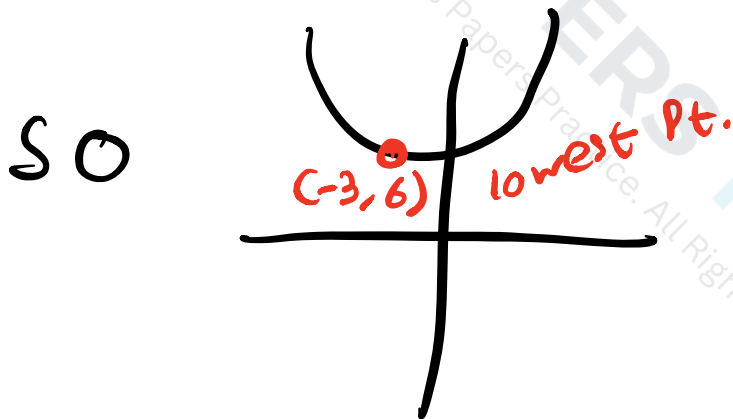
$$x^2 - 6x - 5 \geq 2x^2 + 10$$

$$x^2 + 6x + 15 \leq 0$$

roots
 $(x+3)^2 + 6$

$$x = \frac{-6 \pm \sqrt{36 - 60}}{2}$$

$\nexists$
no real roots



so never

There is no x s.t the inequality
is met.

In fact $x^2 + 6x + 15 \geq 6$
for any x